The Journal of Open Source Software

LightLogR: Reproducible analysis of personal light exposure data

Johannes Zauner [1,2¶], **Steffen Hartmeyer** [3], **and Manuel Spitschan** [1,2,4,5]

1 Technical University of Munich, TUM School of Medicine and Health, Department Health and Sports Sciences, Chronobiology & Health, Munich, Germany **2** Max Planck Institute for Biological Cybernetics, Max Planck Research Group Translational Sensory & Circadian Neuroscience, Tübingen, Germany **3** École Polytechnique Fédérale de Lausanne (EPFL), School of Architecture, Civil and Environmental Engineering (ENAC), Laboratory of Integrated Performance in Design (LIPID), Lausanne, Switzerland **4** TUM Institute for Advanced Study (TUM-IAS), Technical University of Munich, Garching, Germany **5** TUMCREATE Ltd., Singapore, Singapore ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Rachel Wegener

Reviewers:

- @cansavvy
- @welch16

Submitted: 07 October 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary

Light plays an important role in human health and well-being, which necessitates the study of the effects of personal light exposure in real-world settings, measured by means of wearable devices. A growing number of studies incorporate these kinds of data to assess associations between light and health outcomes. Yet with few or missing standards, guidelines, and frameworks, setting up measurements, analysing the data, and comparing outcomes between studies is challenging, especially considering the significantly more complex time series data from wearable light loggers compared to controlled stimuli used in laboratory studies. In this paper, we introduce `LightLogR`, a novel resource to facilitate these research efforts in the form of an open-source, permissively MIT-licenced software package for the statistical software R. As part of a developing software ecosystem, `LightLogR` is built with common challenges of current and future datasets in mind. The package standardizes many tasks for importing and processing personal light exposure data, provides quick as well as detailed insights into the datasets through summary and visualization tools, and incorporates major metrics commonly used in the field (61 metrics across 17 metric families), while embracing an inherently hierarchical, participant-based data structure.

Figure 1: LightLogR logo

Statement of need

Personalized luminous exposure data is progressively gaining importance across various domains, including research, occupational affairs, and lifestyle tracking. Data are collected through a proliferating selection of wearable light loggers and dosimeters, varying in size, shape,

functionality, and output format (Hartmeyer et al., 2023). Despite or potentially because of numerous use cases, the field still lacks a unified framework for collecting, validating, and analyzing the accumulated data (Hartmeyer et al., 2023; Spitschan et al., 2022). This issue increases the time and expertise necessary to handle such data and also compromises the FAIRness (findability, accessibility, interoperability, reusability) (Wilkinson et al., 2016) of the results, especially for meta-analyses (Vries et al., 2024).

`LightLogR` (Figure 1) was designed to be used by researchers who deal with personal light exposure data collected from wearable devices. These data are of interest for various disciplines, including chronobiology, sleep research, vision science and epidemiology, as well as for post-occupancy evaluations in architecture and lighting design. The package is intended to streamline the process of importing, processing, and analysing these data in a reproducible and transparent manner. The package is available on GitHub (Zauner, Hartmeyer, et al. (2023b)) and CRAN (Zauner et al. (2024a)), has a dedicated website for documentation and tutorials (Zauner, Hartmeyer, et al. (2023a)), and releases are archived on Zenodo (Zauner et al. (2024b)).

`LightLogR`'s key features include:

- a growing list of supported devices with pre-defined import functions tailored to their data structure (17 at the time of writing, see Table 2),
- preprocessing functions to combine different time series, aggregate and filter data, and find and deal with implicitly missing data,
- visualization functions to quickly explore the data. These function are based on the popular `ggplot2` (Wickham, 2016) plotting package and are designed to be easily customizable to construct publication-ready figures (see, Figure 2),
- a large and growing set of metrics that cover most if not all major approaches found in the literature (at the time of writing 61 metrics across 17 metric families, see Table)), accessible via a consistent function interface.

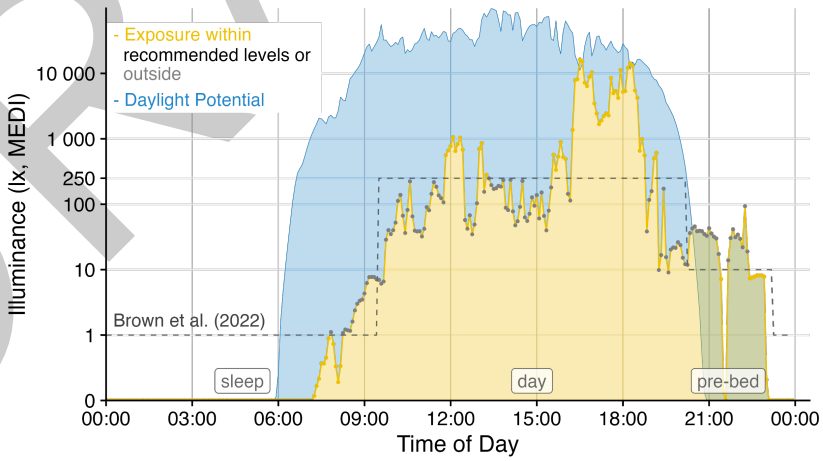

Figure 2: Light logger data can powerfully convey insights into personal light exposure and health-related outcomes. `LightLogR` facilitates the import and combination of different data sources into a coherent data structure, as seen here by combining environmental daylight availability and personal light exposure with data from a sleep diary. The visualization functions in the package further allow customization to produce publication-ready figures. This figure was created with the 'gg_day()' function. The creation process is part of a tutorial (Zauner, Hartmeyer, et al., 2023c) on several key functions in the package.

Device Name	Manufacturer
Actiwatch Spectrum	Philips Respironics

Device Name	Manufacturer
ActLumus	Condor Instruments
ActTrust	Condor Instruments
DeLux	Intelligent Automation Inc.
GENEActiv[1]	Activeinsights
Kronowise	Kronohealth
Lido	Lucerne University of Applied Sciences and Arts
LightWatcher	Object-Tracker
LIMO	École nationale des travaux publics de l'État (ENTPE)
LYS Button	LYS Technologies
Motion Watch 8	CamNtech
melanopiQ Circadian Eye	Max Planck Institute for Biological Cybernetics
XL-500 BLE	NanoLambda
OcuWEAR	Ocutune
Speccy	Monash University Malaysia
SpectraWear	University of Manchester
VEET	Meta Reality Labs

Table 2: Devices supported for import in version 0.4.1

Metric Family	Submetrics	Note	Documentation
Barroso	7		`barroso_lighting_metrics()`
Bright-dark period	4x2	bright / dark	`bright_dark_period()`
Centroid of light exposure	1		`centroidLE()`
Disparity index	1		`disparity_index()`
Duration above threshold	3	above, below, within	`duration_above_threshold()`
Exponential moving average (EMA)	1		`exponential_moving_average()`
Frequency crossing threshold	1		`frequency_crossing_threshold()`
Intradaily Variance (IV)	1		`intradaily_variability()`
Interdaily Stability (IS)	1		`interdaily_stability()`
Midpoint CE (Cumulative Exposure)	1		`midpointCE()`
nvRC (non-visual circadian response)	4		`nvRC()`, `nvRC_circadianDisturbance()`, `nvRC_circadianBias()`, `nvRC_relativeAmplitudeError()`
nvRD (non-visual direct response)	2		`nvRD()`, `nvRD_cumulative_response()`
Period above threshold	3	above, below, within	`period_above_threshold()`

[1]Available after processing of the data using `GGIR` (Migueles et al., 2019).

DRAFT

Metric Family	Submetrics	Note	Documentation
Pulses above threshold	7x3	above, below, within	`pulses_above_threshold()`
Threshold for duration	2	above, below	`threshold_for_duration()`
Timing above threshold (TAT)	3	above, below, within	`timing_above_threshold()`
Total:			
17 families	**61 metrics**		

: metrics available in version 0.4.1

LightLogR is already being used in several research projects across scientific domains, including:

- an ongoing cohort study to collect light exposure data across different geolocations (Guidolin et al., 2024),
- an ongoing cohort study to collect year-long datasets of various types of environmental and behavioral data (Biller et al., 2024),
- a novel power analysis method for personal light exposure data (Zauner, Udovicic, et al., 2023),
- an intervention study on the effects of light on bipolar disorder (Roguski et al., 2024),
- an intervention study on exposure to bright light during afternoon to early evening on later evening melatonin release in adolescents (Lazar et al., 2024),
- an observational study on the wearing compliance of personal light exposure (Stefani et al., 2024),
- an observational study on the differences in light exposure and light exposure related behavior between Malaysia and Switzerland (preregistration in progress),
- an intervention study on sex and seasonal changes in human melatonin suppression and alerting response to moderate light (publication in progress),
- an observational study on light exposure, sleep, and circadian rhythms in hospital shift workers (publication in progress).

Funding Statement

The development of `LightLogR` is funded by MeLiDos, a joint, EURAMET-funded project involving sixteen partners across Europe, aimed at developing a metrology and a standard workflow for wearable light logger data and optical radiation dosimeters (Spitschan et al., 2024). Its primary contributions towards fostering FAIR data include the development of a common file format, robust metadata descriptors, and an accompanying open-source software ecosystem.

The project (22NRM05 MeLiDos) has received funding from the European Partnership on Metrology, co-financed from the European Union's Horizon Europe Research and Innovation Programme and by the Participating States. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or EURAMET. Neither the European Union nor the granting authority can be held responsible for them.

Acknowledgements

We thank Carolina Guidolin (Max Planck Institute for Biological Cybernetics) and Dr. Anna Magdalena Biller (Technical University of Munich) for testing the software during development and providing feature ideas, and the entire Translational Sensory & Circadian Neuroscience Unit (MPS/TUM/TUMCREATE) for its support.

DRAFT

References

Biller, A. M., Fatima, N., Hamberger, C., Hainke, L., Plankl, V., Nadeem, A., Kramer, A., Hecht, M., & Spitschan, M. (2024). The ecology of human sleep (EcoSleep) cohort study: Protocol for a longitudinal repeated measurement burst design study to assess the relationship between sleep determinants and outcomes under real-world conditions across time of year. *Journal of Sleep Research*, e14225. https://doi.org/10.1111/jsr.14225

Guidolin, C., Aerts, S., Agbeshie, G. K., Akuffo, K. O., Aydin, S. N., Baeza Moyano, D., Bolte, J., Broszio, K., Cantarero-García, G., Didikoglu, A., González-Lezcano, R. A., Joosten-Ma, H., Melero-Tur, S., Nilsson Tengelin, M., Pérez Gutiérrez, M. C., Stefani, O., Svensson, I., Udovicic, L., Zauner, J., & Spitschan, M. (2024). Protocol for a prospective, multicentre, cross-sectional cohort study to assess personal light exposure [preprint]. *medRxiv*. https://doi.org/10.1101/2024.02.11.24302663

Hartmeyer, S. L., Webler, F. S., & Andersen, M. (2023). Towards a framework for light-dosimetry studies: Methodological considerations. *Lighting Research & Technology*, *55*(4-5), 377–399. https://doi.org/10.1177/14771535221103258

Lazar, R., Fazlali, F., Dourte, M., Epple, C., Stefani, O., Spitschan, M., & Cajochen, C. (2024). Afternoon to early evening bright light exposure reduces later melatonin production in adolescents [preprint]. *bioRxiv*. https://doi.org/10.1101/2024.10.02.616112

Migueles, J. H., Rowlands, A. V., Huber, F., Sabia, S., & Hees, V. T. van. (2019). GGIR: A research community-driven open source R package for generating physical activity and sleep outcomes from multi-day raw accelerometer data. *Journal for the Measurement of Physical Behavior*, *2*(3). https://doi.org/10.1123/jmpb.2018-0063

Roguski, A., Needham, N., MacGillivray, T., Martinovic, J., Dhillon, B., Riha, R., Armstrong, L., Campbell, I., Ferguson, A., Hilgen, G., Lako, M., Ritter, P., Santhi, N., Schantz, M. von, Spitschan, M., & Smith, D. (2024). Investigating light sensitivity in bipolar disorder (HELIOS-BD). *Wellcome Open Research*, *9*, 64. https://doi.org/10.12688/wellcomeopenres.20557.1

Spitschan, M., Smolders, K., Vandendriessche, B., Bent, B., Bakker, J. P., Rodriguez-Chavez, I. R., & Vetter, C. (2022). Verification, analytical validation and clinical validation (V3) of wearable dosimeters and light loggers. *Digital Health*, *8*, 20552076221144858. https://doi.org/10.1177/20552076221144858

Spitschan, M., Zauner, J., Nilsson Tengelin, M., Bouroussis, C. A., Caspar, P., & Eloi, F. (2024). Illuminating the future of wearable light metrology: Overview of the MeLiDos project. *Measurement*, *235*, 114909. https://doi.org/10.1016/j.measurement.2024.114909

Stefani, O., Marek, R., Schwarz, J., Plate, S., Zauner, J., & Schrader, B. (2024). Wearable light loggers in field conditions: Corneal light characteristics, user compliance and acceptance [preprint]. *Preprints*. https://doi.org/10.20944/preprints202409.1285.v1

Vries, S. de, Gkaintatzi-Masouti, M., Duijnhoven, J. van, Mardaljevic, J., & Aarts, M. (2024). Recommendations for light-dosimetry field studies based on a meta-analysis of personal light levels of office workers. *Lighting Research & Technology*, *55*, 14771535241248540. https://doi.org/10.1177/14771535241248540

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. ISBN: 978-3-319-24277-4

Wilkinson, M. D., Dumontier, M., Aalbersberg, Ij. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., Silva Santos, L. B. da, Bourne, P. E., Bouwman, J., Brookes, A. J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C. T., Finkers, R., … Mons, A. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, *3*, 160018. https://doi.org/10.1038/sdata.2016.18

DRAFT

Zauner, J., Hartmeyer, S. L., & Spitschan, M. (2023a). *LightLogR: Process Data from Wearable Light Loggers and Optical Radiation Dosimeters. Documentation and tutorials*. https://tscnlab.github.io/LightLogR/index.html.

Zauner, J., Hartmeyer, S. L., & Spitschan, M. (2023b). *LightLogR: Process Data from Wearable Light Loggers and Optical Radiation Dosimeters. Github repository* (Version v0.4.1). https://github.com/tscnlab/LightLogR

Zauner, J., Hartmeyer, S. L., & Spitschan, M. (2023c). *LightLogR: Tutorial: The whole game*. https://tscnlab.github.io/LightLogR/articles/Day.html.

Zauner, J., Hartmeyer, S. L., & Spitschan, M. (2024a). *LightLogR: Process Data from Wearable Light Loggers and Optical Radiation Dosimeters. CRAN package*. https://doi.org/10.32614/CRAN.package.LightLogR.

Zauner, J., Hartmeyer, S. L., & Spitschan, M. (2024b). *Tscnlab/LightLogR: v0.4.1. archive* (Version v0.4.1). Zenodo. https://doi.org/10.5281/zenodo.13839724

Zauner, J., Udovicic, L., & Spitschan, M. (2023). Power analysis for personal light exposure measurements and interventions [preprint]. *Research Square*. https://doi.org/10.21203/rs.3.rs-3771881/v1

DRAFT